

Spectrally Filtered Optical Convolutional Neural Network for Shift-Robust Bright-Field Classification

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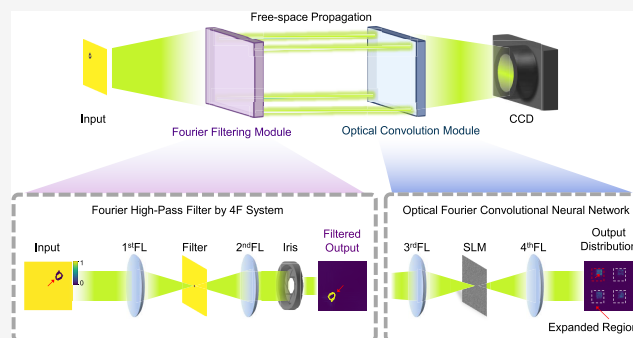
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ABSTRACT: Optical neural networks (ONNs) offer high-throughput and energy-efficient computation, providing a promising alternative to conventional electronic systems. However, their application in bright-field microscopy is limited by random spatial displacements of dynamic targets and a strong unmodulated transmitted-light component concentrated at the zero order of the spatial-frequency domain. This direct current (DC) component significantly reduces feature contrast, posing a major challenge for reliable optical inference. To address these challenges, we propose a spectrally filtered optical convolutional neural network (SF-OCNN) that integrates Fourier filtering, optical convolution, and an expanded detection region. The Fourier filter suppresses the DC component before optical convolution while preserving high-frequency structural information. Optical convolution provides inherent translation equivariance, while the expanded detection region accommodates the displacement of the output response during inference. The proposed system is validated through both numerical simulations and physical experiments. It achieves validation accuracies of 91.12% in simulation and 88.50% in the physical experiment on a bright-field MNIST data set (digits 0–3). For experimentally acquired microfluidic cell images re-encoded on the DMD, the corresponding accuracies are 99.50% and 98.25%. These results demonstrate that combining physical DC suppression with optical convolution provides an effective approach for robust optical computing in realistic bright-field environments.

KEYWORDS: Optical neural networks, Optical convolution, Fourier-domain filtering, Diffractive optical computing, Bright-field microscopy, Translation equivariance



INTRODUCTION

The rapid advancement of deep learning has revolutionized image processing, yet it increasingly exposes the energy and latency bottlenecks of traditional electronic computing hardware.^{1–6} To overcome these limitations, optical neural networks (ONNs) have emerged as a promising paradigm for high-throughput and energy-efficient computing by leveraging the intrinsic properties of light.^{7–10} However, deploying ONNs in practical biomedical imaging scenarios, such as bright-field microscopy and dynamic cell detection,^{11,12} presents unique physical challenges. Unlike idealized laboratory settings, the optical field in bright-field microscopy is inherently dominated by the direct current (DC) component of unmodulated transmitted light, while the dynamic nature of target cells introduces random spatial displacements. The DC component reduces the contrast of structural features, whereas, in position-dependent optical classifiers, spatial displacements alter the interaction of the input field with fixed optical transformations and detector regions. These two effects hinder reliable target recognition over large fields of view.

Among various optical computing architectures, diffractive deep neural networks (D²NNs) have been extensively explored for applications ranging from logic operations to computational imaging.^{13–19} Significant progress has also been made in enhancing their integration density^{20–22} and physical representational capabilities.^{23–28} Conventional D²NNs employ spatially fixed diffractive masks and fixed detector regions, making the learned optical mapping dependent on the absolute input position and therefore not translation equivariant.²⁹ Consequently, spatial displacements reshape the output intensity distribution and redistribute optical energy away from the correct detector region, thereby reducing classification accuracy. Training with center-aligned inputs further limits generalization to unseen positions, while shift

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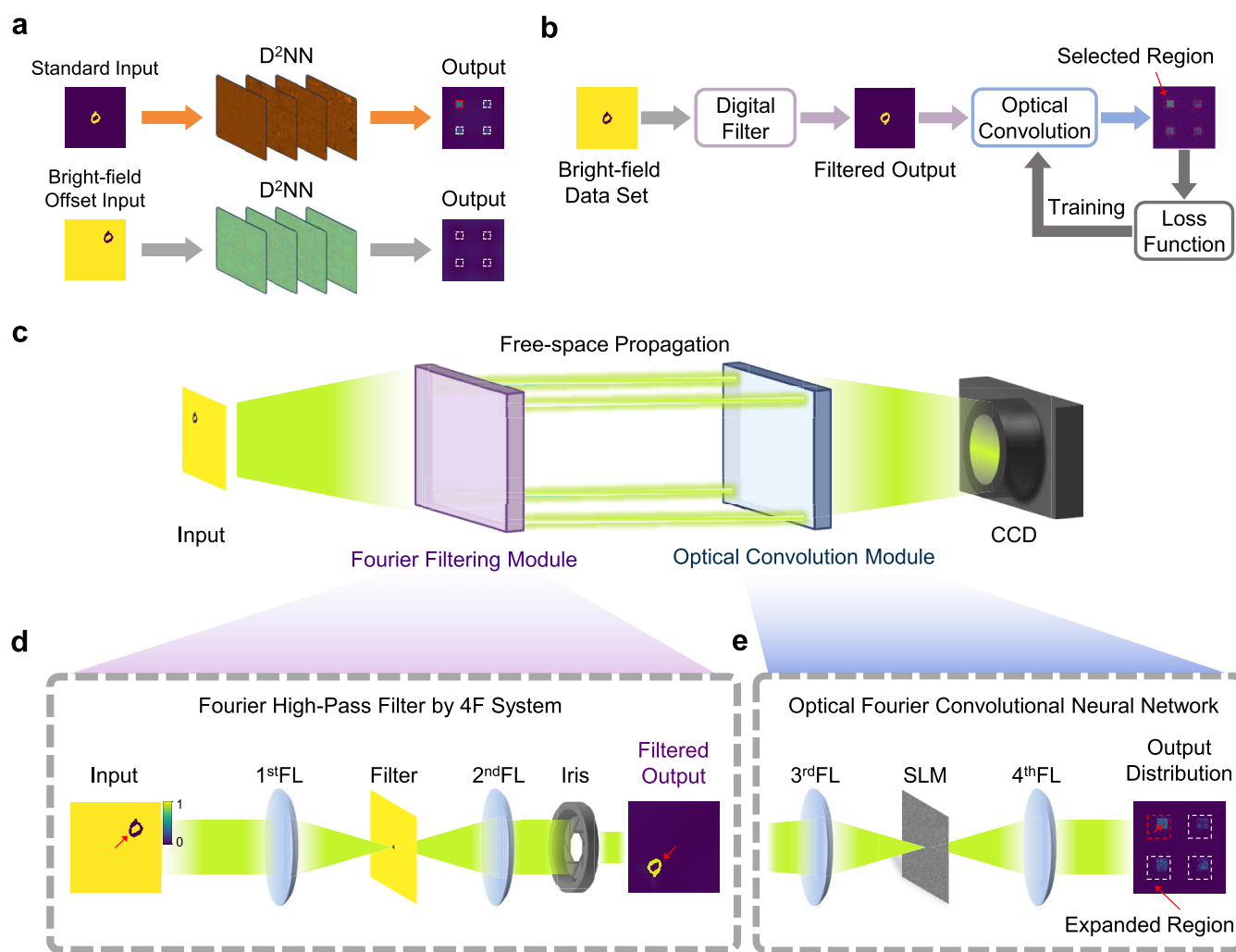


Figure 1. Architecture of the SF-OCNN. (a) Limitations of conventional D²NNs under spatial offsets. (b) Training workflow using centrally aligned images. (c) Overall system layout consisting of Fourier filtering and optical convolution stages. (d) Fourier filtering module: a 4f system with a high-pass mask for DC suppression and feature-contrast enhancement. (e) Optical convolution module: a phase-only SLM performs Fourier-domain convolution, featuring an expanded detection region for shift-robustness.

augmentation provides only data-driven compensation and cannot overcome this architectural limitation.³⁰ In bright-field imaging, the DC component further degrades feature contrast, which shift augmentation cannot restore.³¹

To improve robustness to spatial variations, optical convolutional neural networks (OCNNs) have emerged as a promising alternative. Previous studies have proposed various integrated and 4f-based OCNN architectures for high-speed parallel processing, incorporating techniques such as Fourier-plane spatial filtering and shift-tolerant strategies.^{30,32–39} However, these methods have been primarily validated using high-contrast inputs, where the DC component is weak or negligible. The role of physical DC suppression in enabling robust optical convolution under DC-dominated bright-field conditions has received little attention.

Therefore, we developed a spectrally filtered optical convolutional neural network (SF-OCNN) for robust optical inference under DC-dominated bright-field conditions, as illustrated in Figure 1. The proposed framework integrates two complementary physical mechanisms. First, a spatial high-pass filter is introduced at the Fourier plane to suppress the DC component before optical convolution, thereby enhancing the contrast of informative structural features. Second, optical

convolution provides translation equivariance for spatially shifted inputs, while the expanded detection region accommodates the corresponding displacement of the output response during inference. The results demonstrate that the proposed method effectively mitigates the impact of the DC component and spatial shifts, establishing a robust and efficient optical computing framework for dynamic target analysis in practical bright-field imaging scenarios.

RESULTS AND DISCUSSION

Optical System Architecture

The proposed SF-OCNN consists of two cascaded sub-modules: the Fourier Filtering Module and the Optical Convolution Module (Figure 1c). The Fourier Filtering Module is based on a standard 4f system (Figure 1d). The input bright-field image is Fourier-transformed by the first lens to generate a spatial frequency distribution at its back focal plane. In bright-field conditions, the optical field is dominated by the DC component. While digital preprocessing can remove this component before convolution, the all-optical processing path following DMD input encoding suppresses it physically at the Fourier plane to preserve weak structural features for

subsequent optical convolution. Additionally, an adjustable iris at the module's output confines the transmitted field to the region containing the filtered target features and suppresses peripheral boundary responses before it enters the next stage.

The Optical Convolution Module is designed for all-optical feature extraction and classification (Figure 1e). Also utilizing a 4f system, this module receives the filtered light field and performs phase modulation via a programmable or prefabricated mask positioned at its Fourier plane. After an inverse Fourier transform, the modulated field produces a spatial intensity distribution at the output plane, where classification is performed by comparing the optical energy collected within the predefined detection regions.

Physical Modeling of Translation Equivariance and Inference Strategy

To theoretically validate the system's robustness against translational shifts in bright-field imagery, we developed a general light-field propagation model based on a coherent 4f optical architecture. Let (x, y) denote the spatial coordinates in the input plane, and $U_{\text{in}}(x, y)$ represent the complex amplitude distribution of the input bright-field image. Considering that samples in bright-field microscopy are typically isolated against a uniform background, we first derive the translation relation under the ideal assumption that the uniform field is spatially unbounded. Under this condition, a physical displacement of the sample, $(\Delta x, \Delta y)$, is represented by the translated field $U_{\text{in}}(x - \Delta x, y - \Delta y)$. The light field propagates through the first lens (focal length f_1) to the Fourier plane. Invoking the Fourier shift theorem, a spatial translation corresponds to a linear phase shift in the frequency domain. The spectral distribution $\tilde{U}_f$ is thus given by

$$\begin{aligned}\tilde{U}_f(f_x, f_y) &= \mathcal{F}\{U_{\text{in}}(x - \Delta x, y - \Delta y)\} \\ &= \tilde{U}_0(f_x, f_y) \cdot \exp[-j2\pi(f_x \Delta x + f_y \Delta y)]\end{aligned}\quad (1)$$

where $\mathcal{F}$ denotes the two-dimensional (2D) Fourier transform operator, and $\tilde{U}_0$ represents the spectrum of the unshifted input. The spatial frequency coordinates (f_x, f_y) relate to the physical Fourier plane coordinates (x_f, y_f) via $f_x = x_f/(\lambda f_1)$ and $f_y = y_f/(\lambda f_1)$, where λ is the operating wavelength. This spectrum is modulated by a stationary optical mask with a transmission function $H(f_x, f_y)$ —representing either the high-pass filter in the filtering module or the phase mask in the convolution module. Since the mask is centered on the optical axis and independent of input shifts, the modulated spectrum becomes

$$\tilde{U}_{\text{mod}} = [\tilde{U}_0(f_x, f_y) \cdot H(f_x, f_y)] \cdot \exp[-j2\pi(f_x \Delta x + f_y \Delta y)] \quad (2)$$

Subsequently, the field undergoes an inverse Fourier transform via the second lens (f_2) onto the output plane. Defining the magnification of the corresponding 4f stage as $M = -f_2/f_1$, and applying 4f imaging theory, the output field distribution $U_{\text{out}}(x', y')$ is derived as

$$\begin{aligned}U_{\text{out}}(x', y') &= \iint_{-\infty}^{+\infty} [\tilde{U}_0(f_x, f_y) \cdot H(f_x, f_y)] \cdot e^{-j2\pi(f_x \Delta x + f_y \Delta y)} \\ &\quad e^{j2\pi(f_x \frac{x'}{M} + f_y \frac{y'}{M})} df_x df_y\end{aligned}\quad (3)$$

By rearranging the phase terms within the integral using exponential rules, we obtain

$$\begin{aligned}U_{\text{out}}(x', y') &= \iint_{-\infty}^{+\infty} [\tilde{U}_0 \cdot H] \cdot \\ &\quad \exp\left\{j2\pi\left[f_x \frac{(x' - M\Delta x)}{M} \right. \right. \\ &\quad \left. \left. + f_y \frac{(y' - M\Delta y)}{M}\right]\right\} df_x df_y\end{aligned}\quad (4)$$

Defining $U_{\text{ref}}(x', y')$ as the system response to a stationary input ($\Delta x = \Delta y = 0$), the integral reduces, under this ideal assumption, to a spatially translated version of the reference response

$$U_{\text{out}}(x', y') = U_{\text{ref}}(x' - M\Delta x, y' - M\Delta y) \quad (5)$$

For the ideal model, eq 5 shows that an input displacement $(\Delta x, \Delta y)$ produces a corresponding geometric shift $(M\Delta x, M\Delta y)$ of the output intensity profile while preserving its relative spatial distribution. In the finite system, practical deviations arise from the fixed boundary of the input field, residual boundary responses transmitted through the iris, and truncation of the shifted field by the iris or CCD. These effects are minimized when the displaced target remains within the uniform input field of view (FOV), the effective translated feature remains within the iris aperture, the output field remains within the recorded CCD area, and the active areas of DMD2 and the SLM cover the Fourier-plane field used by the optical model. Equation 5 can therefore be regarded as a good approximation over the designed displacement range. A detailed analysis of these finite-aperture conditions is provided in the Supporting Information.

By contrast, conventional D²NNs use spatially fixed diffractive masks and detector regions, making their optical transformations dependent on absolute coordinates and therefore not translation equivariant.

The proposed inference framework combines translation equivariance with the Expanded Region to simplify training, as illustrated in Figure 1b. The phase mask is optimized using center-aligned bright-field images, and the output response of each class is confined to a preset Selected Region. Within the designed displacement range, an input displacement translates this learned response without changing its relative spatial distribution, while the Expanded Region captures the translated response during inference. Center-aligned training is therefore sufficient, and training with shifted images or shift-based data augmentation is unnecessary.

Specifically, we employed a weighted composite loss function comprising the Negative Pearson Correlation Coefficient (NPCC) and Cross-Entropy loss. The NPCC term encourages concentration of the output optical energy within the target region and reduces spatially dispersed responses. This localization helps maintain separation between the target response and adjacent detection regions when the Expanded Region is used during shifted inference (details provided in Supporting Information).

During inference, each Expanded Region is centered on the corresponding Selected Region. When all lengths are expressed in the detector-plane coordinates, the side length of a square Expanded Region is chosen to satisfy

$$L_{\text{exp}} \geq L_{\text{sel}} + 2|M_{\text{CCD}}|\Delta_{\text{max}} \quad (6)$$

where L_{sel} and L_{exp} are the side lengths of the Selected and Expanded Regions, respectively, M_{CCD} is the total magnifica-

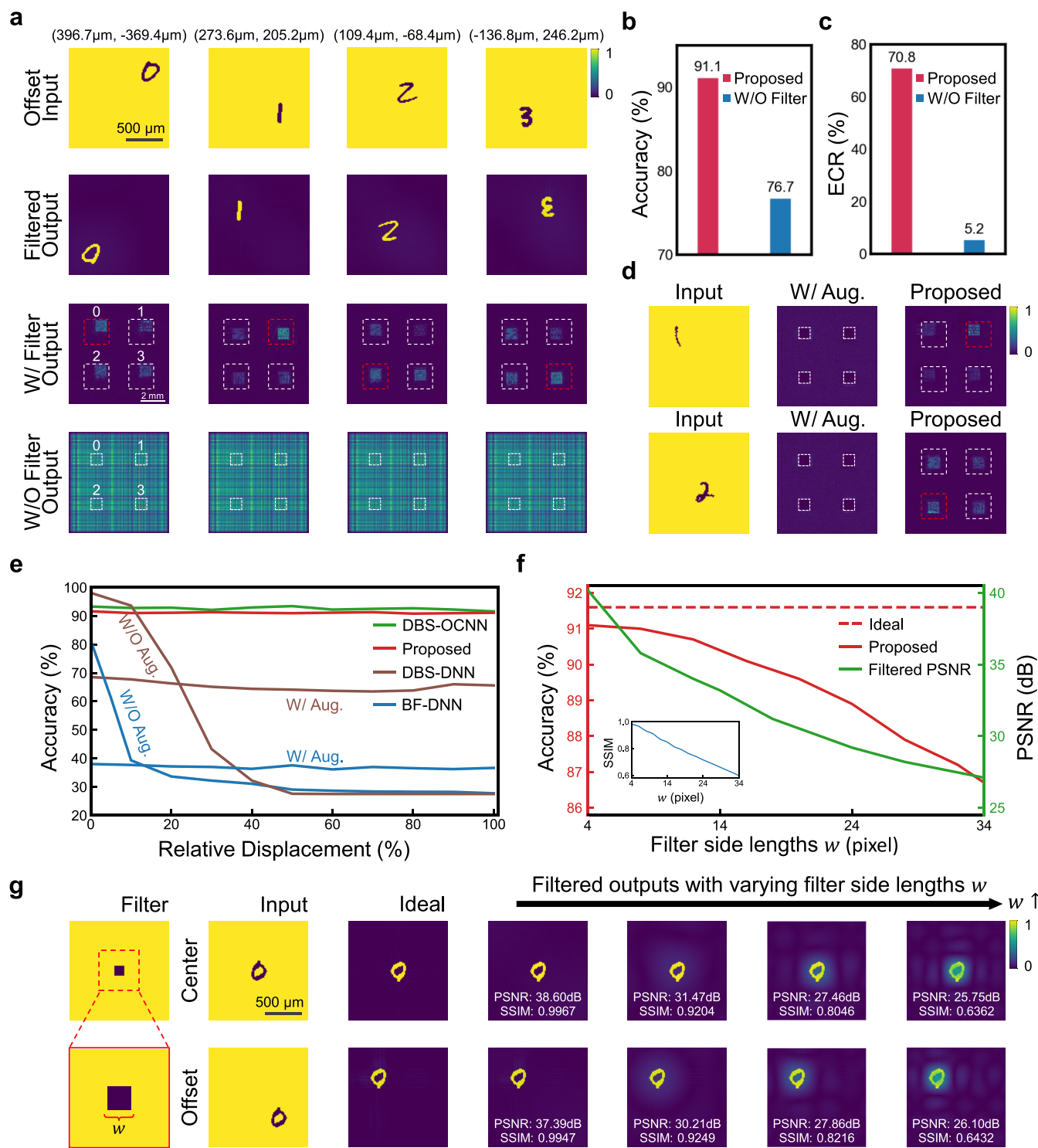


Figure 2. Performance evaluation of the Fourier high-pass filter and comparative analysis of the system. (a) Visualization of signal propagation and DC suppression. (b) Comparison of classification accuracy between the proposed architecture and the filter-less counterpart. (c) Comparison of the ECR. (d) Representative output distributions of the single-layer diffractive neural network for shifted bright-field inputs without and with shift augmentation. (e) Classification accuracy versus relative displacement for the proposed method, digitally background-subtracted OCN, and single-layer diffractive neural networks directly processing shifted bright-field inputs. W/O Aug. and W/ Aug. denote training without and with full-range shift augmentation, respectively. (f) Classification accuracy and validation-set mean PSNR as functions of the central stop width w , with the inset showing validation-set mean SSIM. (g) Representative filtered outputs and corresponding PSNR and SSIM values for centered and shifted targets.

tion from the input plane to the CCD, and $\Delta_{\max}$ is the maximum input displacement along either axis. The two equal-focal-length $4f$ stages used here give $M_{\text{CCD}} = +1$. The Expanded

Region therefore extends the Selected Region by the maximum output displacement on each side. It must also remain within the recorded CCD area without interfering with adjacent class

regions. The corresponding region-sizing criterion is further detailed in the Supporting Information.

Analysis of Spectral Filtering Efficacy and Translation Equivariance

We conducted numerical simulations using a bright-field MNIST data set with random spatial offsets to evaluate the system's performance. All optical propagation simulations were performed on a finite 600×600 -pixel computational field. For a 28×28 -pixel target displaced by up to ± 28 pixels along either axis, the nominal target-support sweep is 84×84 pixels and remains within the input FOV. The SF-OCNN model further included a finite numerical iris after Fourier filtering to transmit the effective target feature throughout this displacement range while suppressing peripheral boundary responses, and the translated output remained within the Expanded Region. In bright-field imagery, the dominant optical energy is concentrated in the DC component. Suppressing this DC component before optical convolution improves feature contrast and classification performance in the SF-OCNN. To assess the impact of the spatial filter, we compared the proposed SF-OCNN against an unfiltered control group. To ensure a rigorous baseline, the control group was trained with comprehensive training-stage data augmentation (100% random spatial offsets).

Despite this algorithmic compensation, the unfiltered system failed to classify shifted bright-field images effectively. As demonstrated in Figure 2a, the high-pass filter suppresses the DC component, isolating discriminative edge features. Consequently, the filtered network precisely concentrates optical energy within the target detection regions, achieving a classification accuracy of 91.1%, compared with 76.7% for the unfiltered system (Figure 2b). To quantify the output energy concentration, the Energy Concentration Ratio (ECR) is defined as

$$\text{ECR} = \frac{\sum_{(x',y') \in \Omega_{\text{tar}}} |U_{\text{out}}(x', y')|^2}{\sum_{(x',y') \in \Omega_{\text{out}}} |U_{\text{out}}(x', y')|^2} \times 100\% \quad (7)$$

where Ω_{tar} denotes the correct-class target detection region and Ω_{out} denotes the complete sampled output plane. The reported ECR values are averaged over the validation set. The filtered network achieves an ECR of 70.8%, compared with only 5.2% for the unfiltered system (Figure 2c). These results indicate that training-stage data augmentation cannot effectively compensate for the loss of feature contrast caused by the DC component and therefore cannot replace physical DC suppression.

Beyond DC suppression, we evaluated the shift robustness of the proposed method against a single-layer diffractive neural network (DNN) and an OCNN under different background-processing conditions. Reference-based digital background subtraction (DBS) was applied to the normalized bright-field inputs before optical inference to establish the DBS-DNN and DBS-OCNN baselines. The background-subtracted input was calculated pixelwise as

$$U_{\text{DBS}}(x, y) = U_{\text{bg}}(x, y) - U_{\text{BF}}(x, y) \quad (8)$$

where U_{BF} is the normalized bright-field input and U_{bg} is the reference field recorded without a target. For the normalized binary inputs used here, $U_{\text{bg}}(x, y) = 1$, and the operation reduces to $U_{\text{DBS}}(x, y) = 1 - U_{\text{BF}}(x, y)$. The DNNs were trained either with center-aligned inputs or with full-range shift

augmentation; the model applied directly to shifted bright-field inputs is denoted BF-DNN in Figure 2e.

For the DNN comparison, the original 28×28 -pixel target was magnified by a factor of 7 to a 196×196 -pixel support within the same 600×600 -pixel computational field. The displacement was scaled by the same factor to preserve the relative shift range, allowing the augmented inputs to sample nearly the entire trainable aperture at 100% displacement. The DNN and OCNN phase masks contained the same number of trainable parameters, and all DNN baselines followed the same training and tuning protocol. The DNN baselines were trained and evaluated using fixed class-specific detector regions, whereas the OCNN-based models used the Expanded Regions defined for translation-equivariant inference. In all cases, class scores were obtained by integrating optical energy within the corresponding detector regions.

As shown in Figure 2d, when the DC component and spatial displacement are simultaneously present, the DNN produces diffuse and incorrectly distributed output responses even after shift augmentation. Figure 2e further shows that, without shift augmentation, the DNN accuracy decreases rapidly with displacement under both background-processing conditions. Full-range shift augmentation stabilizes the accuracy over the evaluated positions, but only at $\sim 37\%$ for BF-DNN and $\sim 65\%$ for DBS-DNN. Data augmentation therefore provides compensation for the sampled positions but does not remove the position dependence of the DNN transformation. The improvement following digital background subtraction shows that electronic removal of the DC component is straightforward and effective, although it introduces an additional preprocessing stage with the associated optoelectronic conversion and computational overhead. Within the optical processing path, the DC component must instead be suppressed physically before convolution.

Because the DNN employs a spatially fixed phase mask and fixed detector regions, its learned transformation is tied to absolute coordinates and is not translation equivariant. This position dependence is an architectural limitation rather than a consequence of the bright-field DC component, and it persists whether the DC component is retained or digitally removed. By contrast, DBS-OCNN maintains an accuracy above 90% across the evaluated displacement range through the translation equivariance of optical convolution. The proposed method achieves similarly stable performance by physically suppressing the DC component before optical convolution, approaching the accuracy of DBS-OCNN while retaining an all-optical feature-extraction path without intermediate electronic preprocessing.

Finally, we evaluated and selected the stop width (w) of the Fourier filter, which affects DC suppression, signal-collection efficiency, and image fidelity (Figure 2f–g). A narrow stop effectively suppresses the DC component while preserving structural integrity; however, excessive stop sizes lead to low-frequency loss and pronounced ringing artifacts. To evaluate DC suppression, PSNR was calculated between the filtered amplitude distribution and the corresponding ideal DC-free reference over the central 100×100 -pixel output region. Both amplitude distributions were independently normalized by their respective peak values to $[0, 1]$. SSIM was additionally calculated within the target-aligned 28×28 -pixel region to characterize the structural similarity between the filtered target feature and its ideal reference. For the representative outputs shown in Figure 2g, PSNR decreases from ~ 38 dB to ~ 26 dB,

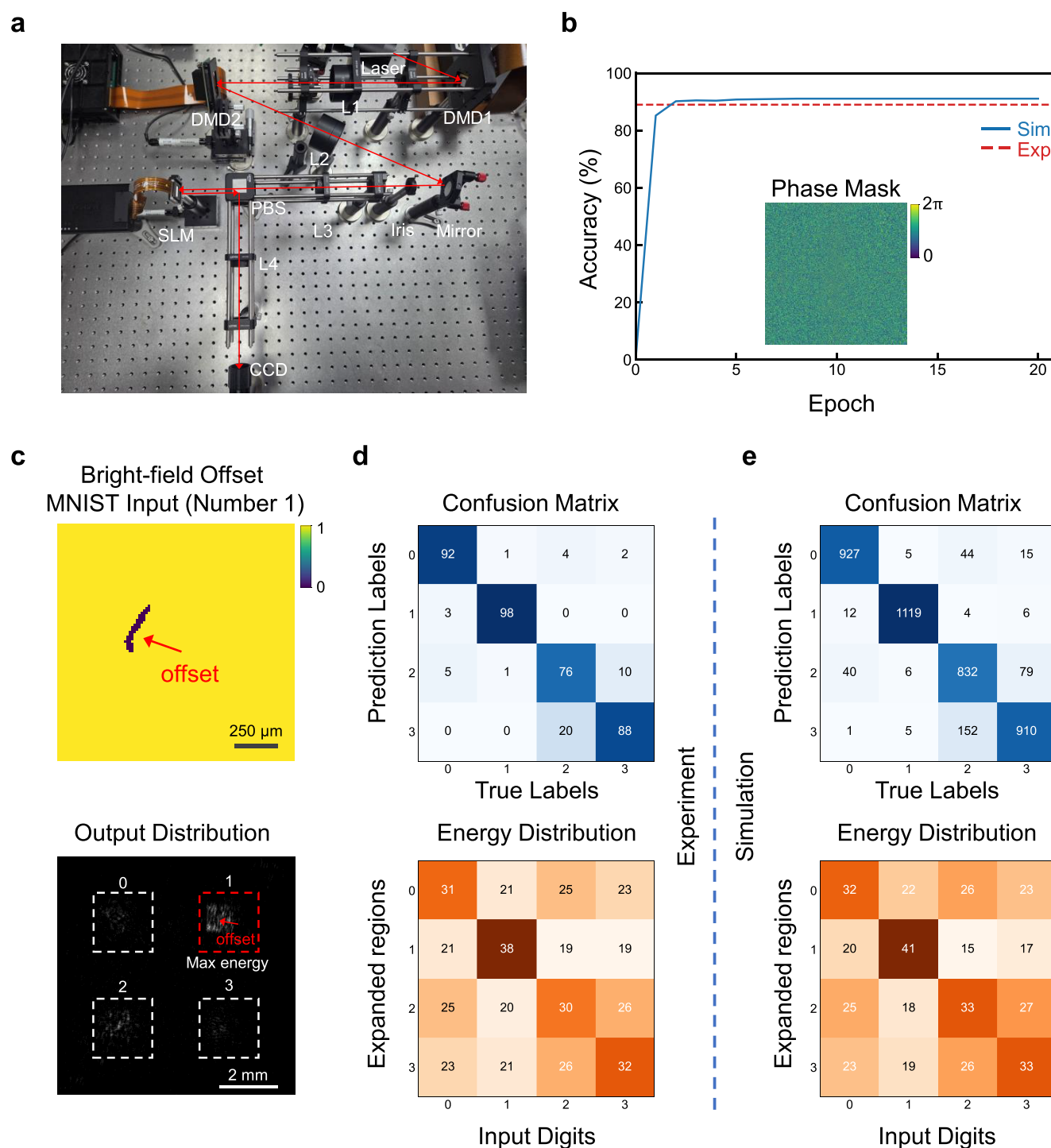


Figure 3. Experimental implementation and performance evaluation on the bright-field MNIST data set. (a) Photograph of the optical setup. (b) Training convergence curve for simulation and achieved experimental accuracy; inset shows the optimized phase mask loaded onto the SLM. (c) Representative result for a shifted bright-field digit “1”. The input image (top) contains spatial offset, while the output field (bottom) concentrates optical energy in the correct target region. (d) Confusion matrix and energy distribution for the physical experiment (400-image validation subset, 88.50% accuracy). (e) Confusion matrix and energy distribution for numerical simulation (4,157 samples, 91.12% validation accuracy).

while SSIM decreases monotonically from ~ 0.997 to ~ 0.636 , as w increases from 4 to 34 pixels. Figure 2f summarizes the corresponding validation-set mean PSNR and SSIM together with the classification accuracy. Both mean metrics exhibit overall decreasing trends with increasing w , while the classification accuracy decreases from $\sim 91\%$ to $\sim 87\%$ over the same range (Figure 2f). Definitions and calculation details for both metrics are provided in the Supporting Information.

Based on our optical parameters, the theoretical minimum stop width required to cover the DC lobe is approximately $25.93 \mu\text{m}$, corresponding to 1.90 DMD pixels. A 2-pixel stop therefore leaves negligible centering tolerance, whereas $w = 4$ provides approximately one DMD pixel of tolerance on each side. At $w = 4$, the filter retains 99.15% of the target-signal energy, corresponding to a loss of only 0.61 percentage points relative to $w = 2$, while providing 38.01 dB of background

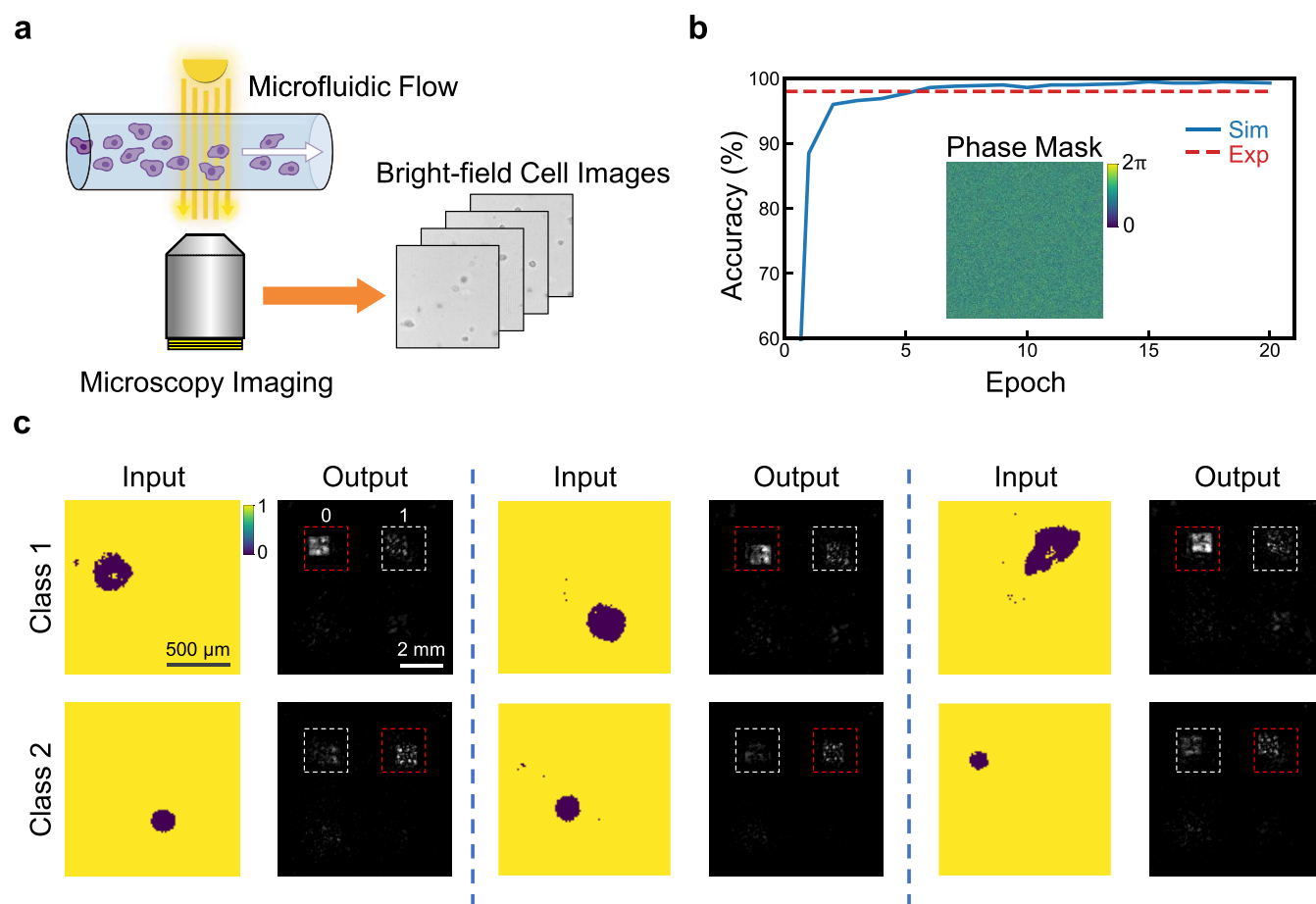


Figure 4. Experimental validation using bright-field cell images acquired from microfluidic flow. (a) Schematic of the data acquisition process, where high-speed flowing cells are captured by bright-field microscopy to generate a data set with inherent random lateral shifts. (b) Training convergence profile and experimental testing accuracy for binary classification. The inset displays the phase mask optimized for cellular features. (c) Representative experimental inference results displaying CTCs (Class 1) in the top row and T cells (Class 2) in the bottom row.

rejection within the output iris (Figure S2 in the Supporting Information). We therefore selected $w = 4$ as the smallest practical stop for all subsequent experiments. The energy definitions, numerical results, and scale-dependent design procedure are provided in the Supporting Information.

Experimental Implementation and Performance Evaluation

The proposed SF-OCNN was implemented in a physical optical system to evaluate its performance under realistic conditions, as shown in Figure 3a. In this setup, bright-field input images with random spatial displacements are displayed on a digital micromirror device (DMD). The optical field is Fourier-transformed to the first Fourier plane, where a second DMD is used as a high-pass filter to suppress the DC component. The filtered signal is then directed to an SLM, which applies the pretrained phase mask for optical convolution. The resulting intensity distribution is finally recorded by a CCD camera at the output plane.

The system was evaluated using a four-class subset of the MNIST data set (digits 0–3). The input images were subjected to random spatial displacements within $\pm 100\%$ of the target region size. As shown in Figure 3b, the proposed system achieves validation accuracies of 91.12% in simulation on 4157 images and 88.50% in the physical experiment on a 400-image subset. Despite the large input displacements, the optical energy in the output plane remains concentrated within

the designated regions, as illustrated by the representative examples in Figure 3c.

The confusion matrices and energy distribution statistics for both simulation and experiment are presented in Figure 3d–e. The performance gap between simulation and experiment is mainly attributed to phase and amplitude errors of the SLM, inaccuracies in the measured propagation distances, and minor system misalignment. Overall, the close agreement between simulation and experimental results verifies the effectiveness of the proposed system in handling bright-field inputs with large spatial shifts.

Classification of Images of Flowing Cells

Optical neural networks offer high-speed and high-throughput computational capabilities, making them promising for biomedical applications such as microfluidic flow cytometry. However, applying conventional ONNs in this context is challenging due to the random lateral displacements of fast-moving cells within wide microfluidic channels, which lead to severe bright-field shift conditions.

Bright-field cell sequences were acquired using a cross-junction microfluidic imaging system. The flow and acquisition conditions are described in Methods section.

To evaluate the robustness of the proposed SF-OCNN under such conditions, we conducted classification experiments using bright-field microscopy images acquired from a microfluidic channel, as shown in Figure 4a. The data set

consists of two representative cell types: circulating tumor cells (CTCs, Class 1) and T cells (Class 2). After acquisition and binarization, these images were re-encoded on the input DMD under coherent 532 nm illumination, with controlled lateral shifts introduced to emulate the spatial variability of flowing cells.

The proposed system achieves a classification accuracy of 99.50% in simulation and 98.25% in physical experiments across 400 test samples, as shown in Figure 4b. Morphologically, CTCs are typically larger and more irregular, whereas T cells are smaller and more circular. The SF-OCNN effectively captures these structural differences while suppressing the influence of the dominant DC component.

Furthermore, the optical convolution combined with spectral filtering supports concentrated output responses within the correct detection regions despite significant spatial displacements, as illustrated in Figure 4c. These results demonstrate that the proposed framework provides a robust and efficient optical computing approach for analyzing dynamically distributed targets under realistic bright-field imaging conditions.

During hardware-synchronized optical inference, the classification accuracy remained at or above 98% as the synchronized input-display and output-acquisition rate increased from 1 to 200 fps (Figure S3 in the Supporting Information). The maximum demonstrated throughput of 200 cell-image inferences per second was set by full-frame CMOS acquisition and hardware synchronization in the current experimental configuration. A representative optical output sequence recorded at 200 fps is provided in Supporting Video S1 at a reduced playback speed, and the associated system-performance estimates are summarized in the Supporting Information. With sufficiently fast region-integrating detectors and matched readout electronics, a higher operating rate could be supported by the binary-pattern refresh of DMD1.

In a direct coherent implementation, the transmitted cell field would contain continuous amplitude and phase variations rather than a binary-amplitude pattern. In this case, Fourier high-pass filtering retains its role in suppressing the zero-order component, while spatially invariant optical convolution preserves translation equivariance for lateral cell displacements. Under partially coherent and incoherent illumination, optical propagation is described through the cross-spectral density or image intensity, respectively.²⁶ For a spatially invariant system, a shift of either input distribution produces the corresponding shift at the output, and translation equivariance is therefore preserved. The influence of illumination coherence is primarily reflected in the Fourier-filtering response: as the spatial coherence decreases, the unmodulated transmitted light extends over a wider angular range, making the suppression achieved by a fixed central mask dependent on both the coherence length and the mask size. In the incoherent limit, the filter acts through the intensity transfer response, and maintaining effective DC suppression would require a corresponding adjustment of the filtering strategy. Accordingly, extension to direct partially coherent or incoherent bright-field illumination would require coherence-aware propagation and training, together with a filter design matched to the illumination conditions.²⁶ Experimental validation under these conditions remains a subject for future investigation.

CONCLUSION

In summary, we propose and experimentally validate a spectrally filtered optical convolutional neural network that integrates Fourier filtering, optical convolution, and an expanded detection region to address the bright-field shift challenge in optical computing. By placing a high-pass filter at the Fourier plane before optical convolution, the proposed framework suppresses the dominant DC component and improves feature contrast at the physical level. Both simulations and experiments show that the system maintains robust performance in the presence of a dominant DC component and large spatial displacements. High classification accuracy is achieved on both the bright-field MNIST data set and experimentally acquired microfluidic cell images re-encoded on the DMD, demonstrating reliable discrimination of different cell types. These results indicate that incorporating task-specific physical mechanisms into optical system design provides an effective approach for handling complex imaging conditions, offering a robust and efficient solution for dynamic target analysis in bright-field microscopy.

METHODS

Experimental System

The experimental optical system was constructed based on a cascaded coherent 4f architecture. The hardware components, arranged sequentially along the optical axis, are as follows: A monochromatic laser (CNI Laser MLL-S-532, 532 nm) was employed as the illumination source. A digital micromirror device (DMD1, Texas Instruments DLP7000, 1024 × 768 pixels, 13.68 μm pitch) was utilized for input image display. The system incorporates four identical Fourier lenses (L1–L4) with a focal length of $f = 200$ mm. At the first Fourier plane between L1 and L2, an identical DMD (DMD2) was deployed as the hardware high-pass filter. At the second Fourier plane between L3 and L4, a pure-phase SLM (HOLOEYE PLUTO-2.1-VIS-016, 1920 × 1080 pixels, 8 μm pitch) was positioned to load the trained network phase mask. Finally, a CCD camera (S496 × 3672 pixels, 2.4 μm pixel size) was located at the output focal plane of L4 to record the intensity distribution.

In the microfluidic experiment, cells were transported through a cross-junction microfluidic chip, with the sample suspension and phosphate-buffered saline (PBS) supplied at 150 and 300 μL h⁻¹, respectively. Bright-field cell sequences were acquired through a 40× objective using an M230 M high-speed camera at 10,000 fps with an exposure time of 20 μs. The acquired images were subsequently used for optical classification.

For the frame-rate evaluation, the CCD at the output plane was replaced by a monochrome CMOS camera (MV-CB016–10UM-C, Sony IMX273) with a manufacturer-specified SNR of 39.9 dB. Hardware triggering synchronized the sequential input display on DMD1 with output acquisition at selected frame rates from 1 to 200 fps, while DMD2 and the convolutional SLM remained fixed during inference. The same ordered test sequence and optical configuration were used at each frame rate, and class scores were obtained by integrating the optical energy within the corresponding detection regions. The maximum tested rate of 200 fps was limited by full-frame CMOS acquisition and hardware synchronization in the current setup rather than by optical propagation.

Data Set Preparation

To comprehensively evaluate the system performance, we constructed two distinct bright-field data sets: a standard digit data set and a biological cell data set. First, for the bright-field MNIST data set (comprising digits 0 to 3), the original images underwent intensity inversion and binarization. Subsequently, the 28 × 28-pixel digit features were embedded into a uniform bright-field input with an FOV of 600 × 600 pixels. The resulting data set comprises 24,754 images from the official MNIST training partition and 4157 images

from the official test partition, with no overlap between the two subsets. Because the latter partition was used to select the composite-loss weights and central-stop width, it served as the validation set in this study. For the physical experiment, 100 images from each digit class were selected from this validation partition and sequentially displayed on the input DMD. Accordingly, the reported experimental accuracy represents performance on a class-balanced subset of the same validation partition rather than on an independent test set. Furthermore, to validate the method's applicability in biomedical tasks, we curated a binary classification data set consisting of CTCs and T cells, with all samples acquired via bright-field microscopy within microfluidic channels. For each cell, one representative image was manually selected and cropped to a 28×28 -pixel region of interest (ROI), followed by the same preprocessing procedure used for the digit data set. The resulting single-cell images were divided at the individual-cell level into 2000 training samples and 400 test samples. To rigorously quantify the translation robustness of the proposed architecture, random spatial displacements were explicitly introduced into the MNIST validation set and biological test set. Specifically, random shifts were applied along both the x and y axes, with offsets ranging within $\pm 100\%$ relative to the target ROI dimensions.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsphotonics.6c01168>.

Composite-loss definition and weight-grid analysis, Fourier-plane central-stop derivation and practical selection, signal-collection efficiency and background rejection, filtered-output fidelity metrics, finite-aperture conditions for translation-equivariant inference, frame-rate-dependent classification performance, and dynamic system characterization (PDF)

Representative synchronized optical output sequence recorded at 200 fps and displayed at reduced playback speed (MP4)

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